

Ionic-liquid materials for the electrochemical challenges of the future

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Ionic liquids are room-temperature molten salts, composed mostly of organic ions that may undergo almost unlimited structural variations. This review covers the newest aspects of ionic liquids in applications where their ion conductivity is exploited; as electrochemical solvents for metal/semiconductor electrodeposition, and as batteries and fuel cells where conventional media, organic solvents (in batteries) or water (in polymer-electrolyte-membrane fuel cells), fail. Biology and biomimetic processes in ionic liquids are also discussed. In these decidedly different materials, some enzymes show activity that is not exhibited in more traditional systems, creating huge potential for bioinspired catalysis and biofuel cells. Our goal in this review is to survey the recent key developments and issues within ionic-liquid research in these areas. As well as informing materials scientists, we hope to generate interest in the wider community and encourage others to make use of ionic liquids in tackling scientific challenges.

It was with simple triethylammonium nitrate that a pure low-melting salt — an ionic liquid — was first identified more than a century ago. In the 1930s, a patent application described cellulose dissolution using a molten pyridinium salt above 130 °C. It was the need for a sturdy medium for nuclear fuel reprocessing that prompted the study of low-melting-point chloroaluminates. Among the ‘onium’ cations with positive nitrogen(s), those derived from the imidazolium ring proved to be the best choice in terms of melting points and electrochemical stability.

At roughly the same time, the need for new anions for organic polymer electrolytes based on polyethylene oxide led to the concept of a ‘plasticizing anion’, that is, an anion having a delocalized charge and multiple conformations differing only marginally in energy. The archetype of such anions is the bis(trifluoromethylsulphonyl)amide ($\text{CF}_3\text{SO}_2\text{--N--SO}_2\text{CF}_3$) ion, also known as NTf_2^- , in which the extremely electron-withdrawing $\text{CF}_3\text{SO}_2\text{--}$ groups are conjugated and linked by flexible S–N–S bonds. When combined with an imidazolium cation, such as the ethylmethylimidazolium cation, this anion produces a fluid ionic liquid (melting point -15°C) with an ion conductivity comparable to that of the best organic electrolyte solutions; it shows no decomposition or significant vapour pressure up to $\sim 300\text{--}400^\circ\text{C}$ (ref. 1). Surprisingly, it is not miscible with water ($\sim 1,000$ p.p.m. in equilibrium with liquid H_2O), and thus defies the conventional wisdom that states polarity is synonymous with hydrophilicity.

Ionic liquids (Box 1) then developed rapidly, with a reinvestigation of ions, for example quaternary ammonium cations, that had been avoided previously by organic chemists because of unsymmetrical shapes that hindered easy purification through crystallization. The organic chemistry community had earlier engaged in research of media with controllable Lewis acidity (chloroaluminate ionic liquids), but the modern era of ionic liquids has produced numerous ‘neutral’ ionic liquids, that is, those based on ions which are unreactive towards acids or bases, be they Lewis or Brønsted. As a result, it is now difficult to name an organic reaction that has not been performed successfully in these potentially ‘green’ solvents, which can be recycled almost indefinitely with no or minimal use of volatile

organic compounds. Most products made in ionic liquids can be distilled off, in the case of small molecules, or extracted with water or hydrocarbon solvents, at least one of which is usually immiscible with the ionic liquid.

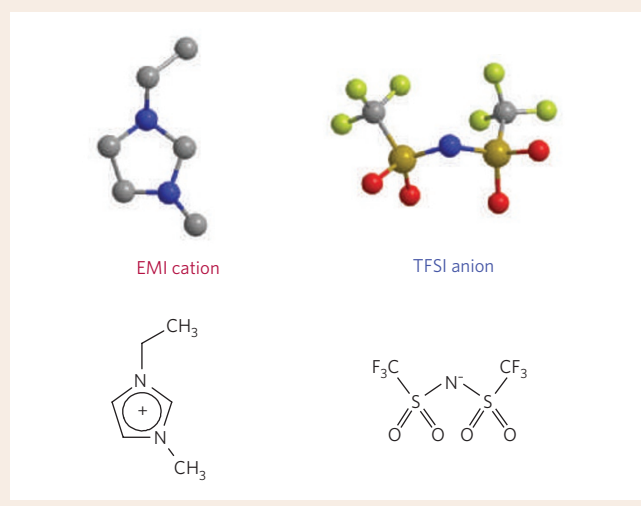
It is this unique solvent potential that makes ionic liquids key materials for the development of a range of emerging technologies. The advent of ionic liquids has made viable processes that fail, or are even impossible, with conventional solvents. Water-sensitive metals or semiconductors that previously could not be deposited from conventional water baths can now, by turning to ionic liquids, be directly electroplated. Energy devices, such as the polymer-electrolyte-membrane fuel cells, lithium batteries and supercapacitors presently under development to address the challenges of increasing energy costs and global warming, may greatly benefit from a switch to low-vapour-pressure, non-flammable, ionic-liquid-based electrolytes. The significant dissolution power of ionic liquids also extends to macromolecules, as best illustrated by the solutions of up to 25 wt% cellulose that can be obtained; this has great consequences for the textile industry and for the only ethically acceptable source of biofuels, polysides from wood. Until it was realized that cellulose was soluble in ionic liquids, the only known solvent for cellulose was an explosive organic compound. Ionic liquids also have applications in biology and biomimetic processes, as their enzymes can be remarkably active, opening a vast field for bioinspired catalysis and biofuel cells.

The scientific and technological importance of ionic liquids today spans a wide range of applications. New types of lubricants and seals, and fluids for thermal engines and adsorption refrigeration have been suggested on the basis of the diverse and unique properties of ionic liquids. We have focused this review on some of the most significant examples of recent key developments in ionic liquid applications made possible by their unique solvent capabilities (Fig. 1 and Box 1). These include electrodeposition, energy management, bioscience and biomechanics. Our main objective is to show how the progress recently achieved in these crucial fields could not have been possible without the advent of ionic liquids.

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Box 1 | Structure of the archetype ionic liquid.

Ionic liquids are low-temperature molten salts, that is, liquids composed of ions only. The salts are characterized by weak interactions, owing to the combination of a large cation and a charge-delocalized anion. This results in a low tendency to crystallize due to flexibility (anion) and dissymmetry (cation). The archetype of ionic liquids is formed by the combination of a 1-ethyl-3-methylimidazolium (EMI) cation and an *N,N*-bis(trifluoromethane)sulphonamide (TFSI) anion. This combination gives a fluid with an ion conductivity comparable to many organic electrolyte solutions and an absence of decomposition or significant vapour pressure up to ~300–400 °C. Ionic liquids are basically composed of organic ions that may undergo almost unlimited structural variations because of the easy preparation of a large variety of their components. Thus, various kinds of salts can be used to design the ionic liquid that has the desired properties for a given application. These include, among others, imidazolium, pyrrolidinium and quaternary ammonium salts as cations and bis(trifluoromethanesulphonyl)imide, bis(fluorosulphonyl)imide and hexafluorophosphate as anions.



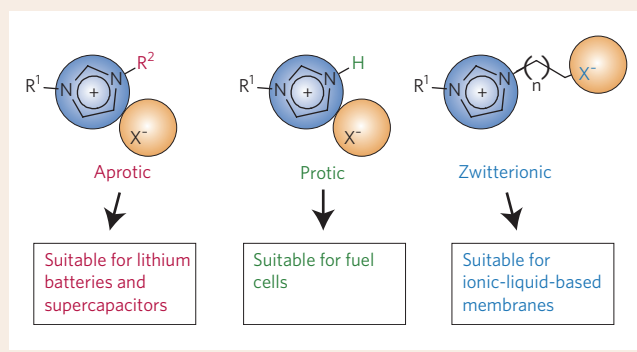
Electrodeposition

Ionic liquids are superior media for the electrodeposition of metals and semiconductors, and have an unprecedented potential to revolutionize electroplating. Key advantages that enable them to overcome the limits imposed by common aqueous or organic media are their wide electrochemical window, spanning up to 6 V in some cases; extremely low vapour pressures, which allow deposition at temperatures well above 100 °C; and numerous, only partly understood, cation/anion effects that make it possible to influence the morphology and crystal size of deposits^{2–4}.

Processes that are impossible in water baths become viable if ionic liquids are used. These include the direct electroplating of water-sensitive metals such as aluminium and possibly even magnesium, as well as many other metals having a deposition potential conflicting with water decomposition. Aluminium is an important metal for corrosion protection of steel as, in contrast to industrially used zinc coatings, it is self-passivating in air and is much more sustainable in terms of known ore resources. However, with an electrode potential of –1.7 V relative to the normal hydrogen electrode, it cannot be deposited from aqueous media, and the well-known SIGAL process^{5,6} involving alkylaluminium compounds in aromatic solvents is hazardous owing to the high flammability of the aluminium precursors used, requiring the strict exclusion of oxygen. The electrodeposition of aluminium from ionic liquids has

Box 2 | Designs of ionic liquids for specific applications.

On the basis of their composition, ionic liquids come in different classes that basically include aprotic, protic and zwitterionic types, each one suitable for a specific application.



Ionic liquids

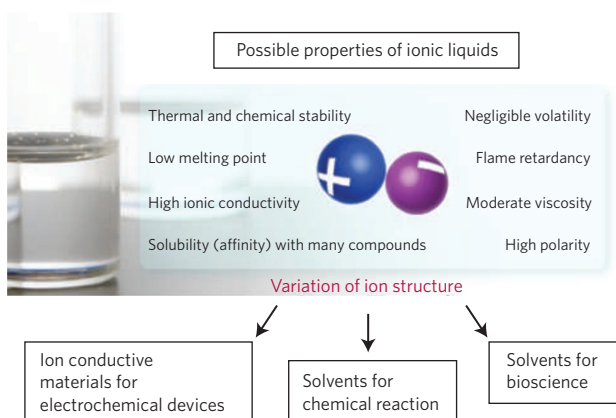


Figure 1 | Ionic liquids, that is, salts with melting points below 100 °C, are potential candidates to be new ‘green’ reaction media with a number of important properties. They can be exploited in various applications, which range from energy storage and conversion to metal deposition, and can be used as reaction media and in chemistry, biochemistry and even biomechanics.

been discussed several times in the literature^{7–9} and is sufficiently advanced that it could be introduced to industry within the next five years as a commercial process. Nonetheless, although ionic liquids are air stable and some are hydrophobic, the presence of AlCl_3 solute species in the ionic-liquid solution as aluminium precursors still requires the exclusion of moisture, and this is a challenge for the next generation of aluminium-plating baths. AlF_4^- , which is less prone to releasing HF than AlCl_3 is to releasing HCl, would be a good choice; however, it disproportionates into AlF_6^{3-} at room temperature. Stabilizing such species in ionic liquids would be very promising.

This process has shown an interesting cation effect: in one ionic liquid, the deposited aluminium is nanocrystalline, with grain sizes of around 50 nm; with a different cation the grain size is, under practically the same conditions, around 5 μm (ref. 10). It is remarkable that just variation of the ionic liquid leads to different morphologies, without the addition of any brightener or grain refiner. This effect results from the solvation layers forming at the interface between the electrode and the ionic liquid; these vary from liquid to liquid and depend upon the metal salt dissolved. Ionic liquids have thus been proven to be excellent media for deposition of a range of

other metals including manganese, nickel, tin and copper; these and presumably many others can easily be deposited from liquids with the dicyanamide anion $(\text{CN})_2\text{N}^-$, which combines charge delocalization and stability at reducing potentials with an ability to complex the metal cation and therefore dissolve and stabilize it¹¹. A further advantage over aqueous plating baths is that many ionic liquids are benign media without toxicity issues in comparison with the cyanides that are involved in many aqueous systems (for example $[\text{Ag}(\text{CN})_2]^-$, which is used in silver plating).

It could be argued that, owing to the very wide electrochemical windows, it should be trivial to deposit reactive elements such as titanium, tantalum, molybdenum and other transition metals that, from the thermodynamic point of view, should be reduced at electrode potentials similar to that of aluminium or even less negative. The main difficulty with these transition elements is the multiplicity of redox states resulting in the electrochemical equivalent of short circuits. From the thermodynamic point of view, the electrochemical window of many ionic liquids is wide enough to allow the deposition of many reactive elements including titanium (from titanium(II)), which lies very close to the deposition potential for lithium. The electrodeposition of thin tantalum layers is possible within a narrow set of experimental parameters, and it has thus been successfully deposited from ionic liquids at elevated temperature, for example to coat biomedical implants^{12,13}. On the other hand, attempts to deposit titanium from titanium halides have hitherto been unsuccessful owing to the formation of titanium subvalent halides, which redissolve into the electrolyte, instead of elemental titanium¹⁴.

As a proof of principle, it was shown that nanoparticles of chromium, molybdenum and tungsten could be made from their respective carbonyl compounds as the source of metal in the bath¹⁵. Once again, a dependence of the particle size on the type of ionic liquid was observed. It seems to be much easier to deposit these elements as alloys; accordingly, the deposition of a ternary aluminium–molybdenum–titanium alloy has recently been described¹⁶.

Ionic liquids also have the potential to be the basis of important approaches in nanoscience. There is great interest in making well-defined metal nanowires, for example using template-assisted techniques. Iron, cobalt and nickel nanowires are, owing to their ferromagnetic properties, interesting for fundamental research. These elements can be obtained in aqueous solution, but as a result of hydrogen coevolution the nanowires are usually of limited quality. This problem does not exist in ionic liquids, as there is no hydrogen evolution, and the template-assisted deposition of silver¹⁷, cobalt¹⁸, germanium and silicon¹⁹ has been reported to deliver compact nanowire mats of these metals; see Fig. 2 for a typical example.

Furthermore, ionic liquids are of importance in semiconductor electrodeposition; recently it was shown that photoluminescent $\text{Si}_x\text{Ge}_{1-x}$ with a bandgap of at least 1.5–3.2 eV can be made by electrodeposition from an ultrapure ionic liquid containing silicon and germanium halides (Fig. 3). During deposition, different colours, from orange to green, are observed in the visible spectrum; these are due to a quantum size effect of the semiconductor particles with sizes between 2 and 20 nm (ref. 20). These results show that the ionic liquid allows deposits so pure that photoluminescence effects can be seen. In addition, the material absorbs visible light and may open the way to a simple electrochemical fabrication of inexpensive solar cells.

However, although the feasibility of metal and semiconductor deposition from ionic liquids has been demonstrated in a variety of cases, there are some fundamental issues that still have to be clarified. The solubility of precursor compounds departs appreciably from that in conventional solvents. The basic electrochemistry of the processes is yet to be fully understood and the structure of the double layer, which, owing to the formation of solvation layers, can obviously not be well described with the models developed for aqueous solutions, must be clarified. The role of the large ions in influencing

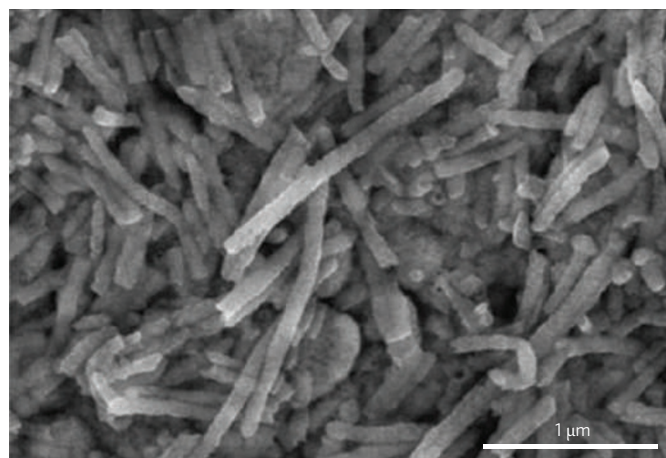


Figure 2 | Processes that are impossible in water baths become viable if an ionic liquid solvent is used. The use of ionic liquid baths also improves the quality and the morphology of the deposits. The figure shows a high-resolution scanning electron microscope image of germanium nanowires made by electrodeposition from ionic liquids. Figure reprinted from ref. 19, © 2008 RSC.

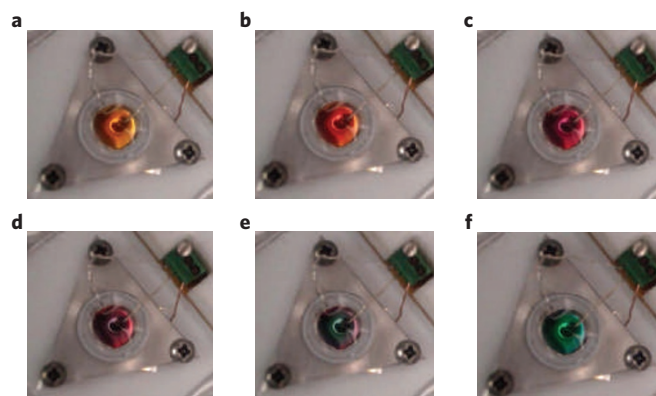


Figure 3 | Deposits of photoluminescent semiconductors, for example $\text{Ge}_x\text{Si}_{1-x}$, obtained from ionic liquid baths. Different colours are seen in the visible spectrum, owing to quantum size effects of the semiconductor particles. This is a clear demonstration that ionic liquids allow deposits of particle size in the nanometre range that are pure enough to make visible the photoluminescence effect. The panels correspond to potentials of –2.62 V (a), –2.68 V (b), –2.74 V (c), –2.78 V (d), –2.83 V (e), and –2.85 V (f). Figure reprinted from ref. 20, © 2008 RSC.

the diffusion kinetics and the mechanism of deposit nucleation and growth are other aspects requiring further investigation.

Energy management

Ionic liquids are also likely to be important in the field of energy. Global concerns about present energy policy, which relies on fossil fuels and suffers the associated economical and ecological problems, call for the increased use of renewable energy sources in households and industry and for the large-scale replacement of internal-combustion-engine vehicles by those with zero or low emissions. This in turn requires the availability of suitable systems both to store the energy in power stations harnessing intermittent sources (solar or wind) and to power electric or hybrid vehicles. Electrochemical systems, such as batteries, are ideal for this purpose²¹. In particular, lithium batteries, which have the highest energy efficiency of all known electrochemical storage systems and which have reached an advanced state of technological development, seem to be the power source of choice. Lithium batteries can have

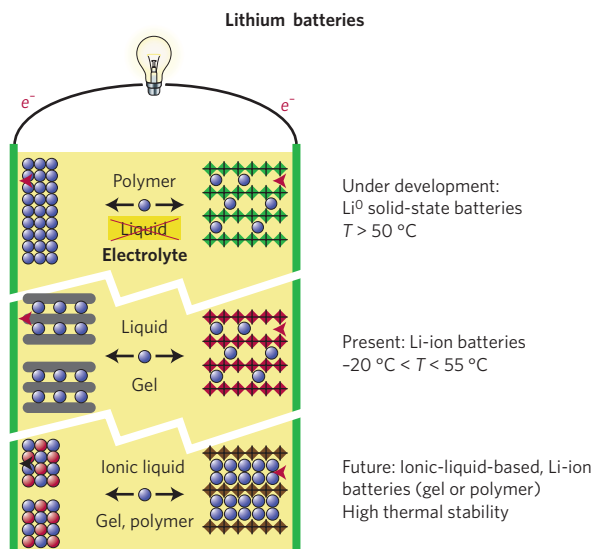


Figure 4 | In their basic structure, lithium batteries are formed by two lithium-exchanging electrodes separated by a lithium-ion-conducting electrolyte. The most common version uses an electrode combination based on a graphite negative electrode with a lithium cobalt oxide positive electrode connected either by a liquid solution of a lithium salt in an organic solvent mixture or by the same solution trapped in a gel. The electrochemical process is the cycling of lithium ions from one electrode to the other. These lithium-ion batteries are commercially available and dominate the consumer-electronics market. However, their use in more demanding applications, such as in sustainable transport, is prevented by a number of drawbacks, among which safety is the main concern. Safety can be improved by excluding liquids, that is, by using solvent-free, polymer electrolytes. The safe operation of this solid-state version, which replaces the graphite with lithium metal at the negative side, has been demonstrated. However, these batteries operate well only at temperatures higher than 50 °C, which limits their range of application. The real breakthrough in the field may come by using ionic-liquid-based gel or polymer electrolytes in combination with revolutionary electrode materials: the high thermal stability and low vapour pressure of the ionic liquids ensure reliability by preventing thermal runaway and pressure build-up, and the innovative electrode structure is expected to enhance the capacity and energy density of the battery.

different configurations (Fig. 4). However, their implementation for energy storage and in vehicle applications has been slowed by safety concerns surrounding the use of large-scale lithium cells. Undesired reactions between the battery components and the liquid organic electrolyte, triggered by unpredictable events such as short circuits or local overheating, lead to an exothermic reaction of the electrolyte with the electrode materials, producing a rapid increase of the battery temperature and, eventually, to fire or explosion.

Again, the unique properties of ionic liquids may help to solve the problem as they are practically non-flammable, which is a significant safety asset. The replacement of the conventional, flammable and volatile, organic solutions with ionic-liquid-based, lithium-ion-conducting electrolytes may greatly reduce, if not prevent, the risk of thermal runaway. This provides the lithium battery with the level of safety that is required for their large-scale application in important and strategic markets. Extensive work in this direction is in progress, and the testing of ionic liquids as new electrolyte media for future, highly safe, lithium batteries is under way in many industrial and academic laboratories^{22–25}. However, the results vary widely, as can be expected from the wide variety of ionic liquids investigated. Key aspects to be understood are, among others, the thermodynamic and kinetic stabilities of the ionic liquids with respect to the electrode materials. It would be naive to expect the

organic ionic-liquid cations to be stable at the extremely negative potentials of alkali-metal deposition; in other words, the average electronegativity of an ion based on carbon, hydrogen or nitrogen cannot be as low that of Li^+ or K^+ . However, the kinetic metastability of the cation and/or protective, passivating-layer formation on the electrode have been welcome observations²⁶. The structure of the interfaces and the nature of the electrode-protecting films are other aspects that need particular attention.

These aspects will be analysed in this review to evaluate their effective influence on the progress of lithium battery technology. The electrolytes that are under study are solutions of a lithium salt in an ionic liquid. The high solubility of lithium salts, even though it requires low-lattice-energy salts (for example LiBF_4 or $\text{Li}[\text{NTf}_2]$), is quite surprising as, in contrast to organic polar solvents, ionic liquids do not possess free electron pairs to provide for a solvation interaction with the cation, other than those on the anion. The resultant structures, probably involving polyanionic species such as $[\text{LiX}_n]^{(n-1)-}$ still need to be better understood. In principle, ionic liquids are only metastable at the potential of lithium reduction; the cations, despite often being resonance stabilized, are sensitive to electron injection, with the subsequent formation of a radical, or proton elimination. For instance, the archetypal ionic-liquid cation, ethylmethyldiazolium, possesses an acidic C2 proton that limits its stability below 1.3 V relative to Li^+/Li^0 (–1.7 V in comparison with H^+/H_2). Yet more and more reports seem^{26–28} to show that in half-cells or full batteries, selected ionic-liquid electrolytes behave well, not only with the conventional graphite negative electrode, but also for Li^0 plating. The latter had been abandoned in organic electrolytes because of uneven plating of the metal during cycling (that is, dendrite formation), with short circuits and dramatic runaway reactions. Graphite was thus selected for ‘Li-ion technology’ but is still a weak point in terms of safety, as solvents can forcefully co-intercalate and exfoliate the graphite with large release of heat.

This behaviour does not seem to occur with ionic liquids, as the solvation is provided by anions that have no tendency to co-intercalate at such potentials. The very high-capacity Li_xSi ($0 < x < 4.4$) alloying electrode (capacity reaching 2,000 mA h g^{-1}) also behaves quite well in contact with ionic liquids. The NTf_2 electrolytes have been shown to cycle well with Li^0 (ref. 26), offering a dramatic increase in possible energy densities and a simpler technology for high-performance batteries. The chemical manipulation of the cation (quaternary ammonium instead of ethylmethyldiazolium) and of the anion $[(\text{FSO}_2)_2\text{N}]^-$ instead of $[\text{CF}_3\text{SO}_2)_2\text{N}]^-$ allows the fine-tuning of the interactions at the interface and passivating-layer formation. Owing to their high anodic stability, ionic liquids function without problem at the positive electrode, even with highly oxidizing materials (for example $\text{LiMn}_{1.5}\text{Ni}_{0.5}\text{O}_4$ and LiMnPO_4). The main problem is technological, as the only acceptable (sustainable) current collector in practical batteries is aluminium, which tends to become corroded at such high voltages. The battery electrolytes based on ionic liquids are thus serious contenders in the design of safe batteries for electric and hybrid vehicles. There is also hope, in the longer term, of building lithium batteries using oxygen from air as the oxidizer at the cathode, which would produce a great increase in capacity, even if the end product of oxygen reduction is Li_2O_2 , that is, only two electrons per O_2 instead of the four possible²⁹. The fact that many of the ionic liquids of interest in this area are hydrophobic is an asset; however, when a lithium salt is dissolved, the electrolyte becomes very water sensitive, and new salts must be designed to counter this problem.

The development of a zero-emission, hydrogen-powered car has been a highly desirable goal for some time. Although the concept is very appealing, the development of hydrogen cars is still restricted to a few models operating in limited geographical regions. The core of the hydrogen car is the hydrogen fuel cell. Fuel cells have been known for more than a century, but some of the issues that have

so far hindered their commercial production are still unsolved³⁰. These are mainly related to limited functional characteristics of the perfluorosulphonic-acid-membrane electrolyte, the conductivity of which is strongly humidity dependent; consequently, its operating temperature is limited to roughly 100 °C (ref. 31). These issues may be addressed to an extent by switching to ionic-liquid-based polymer membranes. This strategy has been tested by immobilizing ionic liquids, by polymerization of the components or gelification by a neutral macromolecule with an architecture allowing a water-independent proton conductivity associated with a high thermal stability^{32–35} (Fig. 5).

In this case, the proton relay molecule, instead of being H₂O, is an imidazole or an amine. Not only is the operating temperature higher, but also the pH increases from 7 to 11. Neutral-to-basic pHs may open new possibilities in terms of catalyst designs, the media being far less aggressive than the highly acidic perfluorosulphonic membrane, with the ultimate goal of replacing platinum. Catalysts made from platinum are considered to be a major barrier to the ultimate widespread uptake of this technology, because of its cost and scarcity. The investigation of ionic-liquid membranes is still in its infancy, but their potential in important, advanced technologies justifies the great attention devoted to them, and this makes it possible to foresee important developments in the near future.

The proper development of electric — or hybrid — vehicle technology requires a complementary component to assist the battery during moments of high power demand such as start-up and acceleration. Supercapacitors — electrochemical devices based on two inert electrodes separated by an electrolyte — are the components of choice³⁶. The challenge in supercapacitor technology is to improve the energy density without sacrificing cycle life and safety. Present supercapacitors are limited to a few watt hours per kilogram, which is well below the specific energy required to provide power assists in hybrid vehicles, and they must be used in tandem with batteries. Hence, structure modifications are required to raise this value, for example by enhancing the voltage of the device. This in turn requires the use of electrolytes that can withstand high voltages without undergoing decomposition. Some limited success has been achieved by passing from aqueous to aprotic solvents, for example acetonitrile or propylene carbonate³⁶. Ionic liquids, owing to their wide electrochemical stability window, are ideal for this application, as demonstrated recently^{37,38}. However, the development of ionic-liquid-based supercapacitors is still in its preliminary stages and further progress is needed to ensure their full exploitation. The challenge is in the choice of ionic liquids that feature wide electrochemical stability windows combined with high ionic conductivity, as well as in the design of electrode structures capable of assuring proper wettability by the ionic liquid.

Bioscience

Bioscience is among the most interesting and important emerging areas where ionic liquids are beginning to play a part. Water has long been believed to be the unique solvent for biomolecules and biosystems. Indeed, almost all studies on enzymatic reactions have been carried out in aqueous media. On the other hand, for long-life biodevices, there are a number of drawbacks to this situation, such as the volatility of water, the limited temperature range and the very narrow pH range for protein stability. Ionic liquids have the potential to overcome these issues. However, to be effective in energy biodevices, such as biofuel cells, ionic liquids must ensure the dissolution of natural biomaterials, as well as producing stable enzymatic reactions.

Dissolution of natural biomaterials, especially of cellulose, is one of the most important targets for energy conversion. Cellulose is produced at a rate of 10¹¹ t yr⁻¹ in nature and, in our opinion, is the only source of chemicals for energy that is ethically acceptable. Recently, large efforts have been devoted to the dissolution of

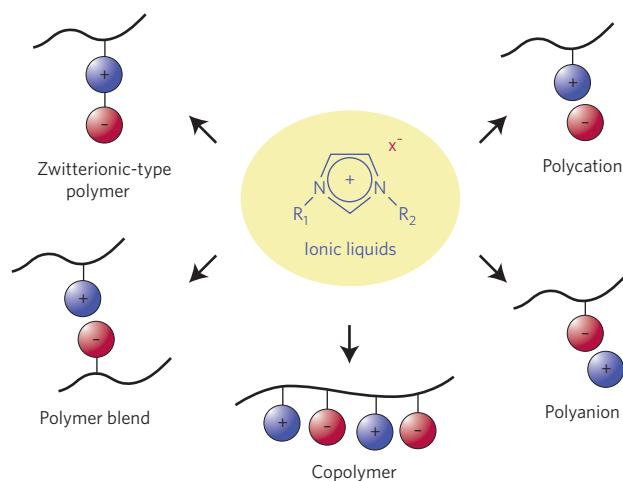


Figure 5 | Although polymer-electrolyte-membrane fuel cells have been known for a long time, they have not yet reached large-scale development as some issues are still unresolved.

These are mainly related to limited functional characteristics of the perfluorosulphonic-acid-membrane electrolyte, whose conductivity relies on the humidity level; consequently, its stability is limited to temperatures below ~100 °C. These issues may be addressed to an extent by switching to ionic-liquid-based polymer membranes. Various strategies have been tested for the preparation of these membranes, including polymerization of the components with the formation of polycations, polyanions, copolymers and special, double-ion structures, as well as blends with a neutral macromolecule, typically polyvinylidene fluoride. These revolutionary architectures offer a water-independent proton conductivity associated with a high thermal stability, which allows high operational temperatures, as well as basic pHs. This opens up new possibilities in terms of catalyst design, with the ultimate goal of replacing platinum, which is impractically rare.

cellulose using polar ionic liquids. Several studies on the preparation of polar and low-melting-point ionic liquids have shown that the key parameter for achieving the dissolution of cellulose is a strong hydrogen-bond-accepting ability of the selected ionic liquid^{39,40}. Initial success was achieved by using 1-butyl-3-methylimidazolium chloride ([bmim]Cl)⁴¹ and 1-allyl-3-methylimidazolium chloride ([amim]Cl)⁴². The important role of the chloride anion is the formation of ionic liquids that show the required hydrogen-bonding ability⁴³. This, however, is counterbalanced by a drawback associated with the melting point of [bmim]Cl (70 °C); hence, extra energy is needed to operate with ionic liquids based on [bmim]Cl. New ionic liquids, in which the phosphate anion derivatives were used instead of the chloride anion, were investigated to solve this issue, and the results demonstrated that such fluids can dissolve cellulose without heating⁴⁴. Thus, owing to these successes, dissolution of cellulose has been achieved; the next step is to preserve its enzymatic reactions in ionic liquids.

The dissolution of enzymes and the maintenance of their effective reactivity require the preservation of their highly ordered structure, a condition that is difficult to achieve even in ordinary ionic liquids, as for the most part they have little ability to solubilize proteins without denaturation. As schematically shown in Fig. 6, there are three different ways to solubilize proteins in ionic liquids. A common method is simply the direct dispersion of the enzymes in ionic liquids. Some lipases are known to be effective in this kind of dispersion. Polyethylene glycol (PEG) modification is the second most generally accepted method for achieving a homogeneous solubilization of proteins in many ionic liquids^{45,46}. PEGylation (that is, chemical modification of PEG chains) is applicable to most proteins and PEGylated proteins are thus soluble in most ionic liquids because of the excellent affinity of PEG chains

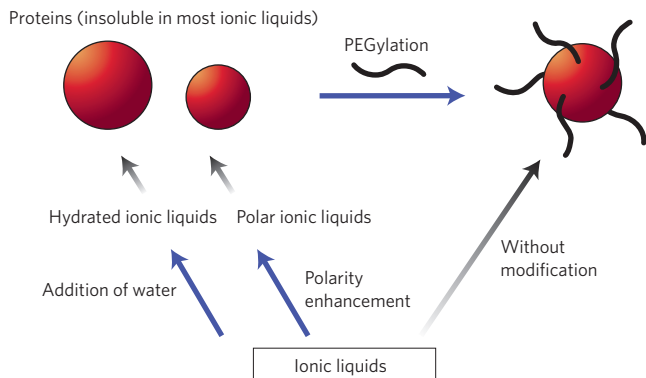


Figure 6 | There are several strategies to solubilize proteins into ionic liquids. A common one is simply the direct dispersion of the enzymes in ionic liquids. Polar ionic liquids are sometimes excellent media in which to solubilize native proteins without any additives or protein modification. Another strategy involves the modification of polyethylene glycol (PEG) on the protein surface, that is, PEGylation. Here, the solubility is improved by the excellent affinity of PEG chains with ions, including those constituting ionic liquids. Finally, mixtures of ionic liquids and a small amount of water, called hydrated ionic liquids, are other potential candidates as solvents for proteins. Because water molecules are strongly bound directly to ions and no free water molecules exist, the solution properties of this hydrated ionic liquid are similar to those of pure ionic liquids.

with ions, including those constituting ionic liquids. Crown ethers sometimes show effects similar to PEG⁴⁷.

A third, more sophisticated approach is based on a relatively new concept. It is not very easy to design ionic liquids having all the desired properties; therefore, an alternative method can be used to achieve the goal of finely controlling the ionic-liquid property, namely that of mixing it with other ionic liquids or even with molecular liquids. Addition of small amounts of water to ionic liquids may strongly influence the protein solubility while retaining the properties of the selected ionic liquid. It is important to note that the chemical activity of water remains very low because all the water molecules are strongly solvated to the ions. Effectively, Fujita and co-workers reported that a water/ionic-liquid mixture, called a hydrated ionic liquid, is an excellent solvent for proteins^{48,49}. Choline dihydrogen phosphate ([Ch][DHP]), composed of a chaotropic cation and a kosmotropic anion, is a typical example. Hydrated (20% water) [Ch][DHP] was shown to be an excellent solvent for cytochrome *c* and other proteins. In this liquid, cytochrome *c* was reported to show electron transfer activity even 18 months after storage at room temperature⁵⁰. It appears that the small amount of water present in the ionic liquid provides the required hydrogen-bonding environment for the protein and hence renders it soluble. Such hydrated ionic liquids are likely to be the solvents of choice for natural biomaterials from now on.

Even in water-free conditions, owing to their strong hydrogen-bonding ability, chloride-based ionic liquids can also act as solvents for native proteins under ambient conditions. For instance, cytochrome *c* was found to be soluble in dry [amim]Cl and [bmim]Cl (ref. 51), showing a quasi-reversible electron transfer reaction at an electrode⁵². Nevertheless, homogeneous dissolution of proteins in ionic liquids remains one of the most critical challenges in this field. In fact, proteins other than cytochrome *c* were found not to be active in dry [amim]Cl and [bmim]Cl, suggesting a unique stabilization of the haem complex with covalent bonds in the globin domain of cytochrome *c*. On the other hand, this also strongly reinforces the idea that, by proper tuning, suitable ionic liquids can be designed to be effective solvents for other proteins.

Following the dissolution of cellulose and enzymes in ionic liquids, the remaining goal is the enzymatic depolymerization of

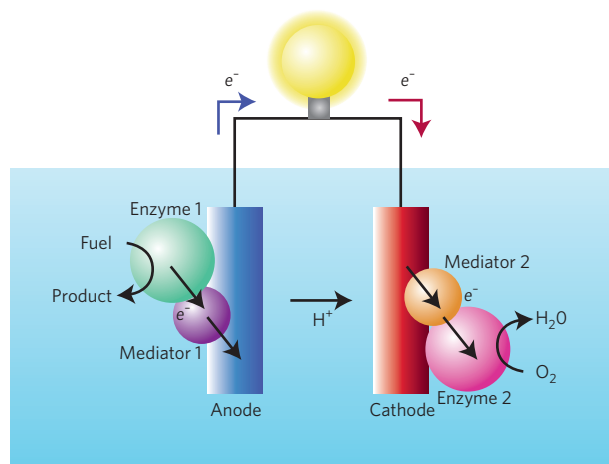


Figure 7 | Sketch for a biofuel cell, that is, a device capable of generating energy with the aid of enzymes. Practical biofuel cells have already been demonstrated using aqueous electrolyte solutions. There is a general opinion that the use of ionic liquids instead of aqueous salt solutions prevents enzymatic activity as a result of the environment that they provide. Considering the large variety of substrates and enzymes available for biofuel cell applications, the use of ionic liquids as electrolytes paves the way for the development of unlimited combinations for novel bio-energy conversion systems, making biofuel cells attractive next-generation energy-conversion devices.

cellulose to generate glucose or other oligosaccharides. Relatively recent reports have described enzymatic hydrolysis in ionic liquid/water mixtures⁵³, which suggests that the goal can be effectively achieved.

Another important goal in the field is the generation of energy from oligosaccharides, cellobiose, glucose or further derivatives. The use of ionic liquids may be of considerable help here also. The generation of electrical energy with the aid of enzymes in a 'biofuel cell' (Fig. 7) has only been developed in aqueous media. However, the above-mentioned issues associated with the use of aqueous media greatly affect the cell operation. Thus, the replacement of water solutions with ionic liquids is expected to enhance the operational life of the biofuel cell, as well as improving its maintenance-free characteristic⁵⁴. However, in this case also a proper selection of the ionic liquid is necessary to achieve valid results. For instance, ionic liquids are required that are good solvents for the (macro)molecular substrate, but that do not support the activity of any virus, algae, fungus, bacteria or similar. On the other hand, many fundamental studies carried out on water-based biofuel cells can be of great help, as they can provide a criterion to be used for the selection of the most suitable ionic liquid to ensure the desired enzymatic energy conversion. Adequately designed ionic liquids should support high solubility and stability of target enzymes as well as preservation of their activity. It is very likely that ionic-liquid-based biofuel cells will be key energy devices of the future.

Biomechanics

Electromechanical actuators, namely devices that show reversible deformation under the stimulus of an applied voltage (Fig. 8), have attracted attention for their biomimetic capability. Among various configurations, those based either on metal-coated ion-exchange polymers or intrinsically electron-conducting polymers seem to be the most promising. In the first case, the device is based on a membrane, coated with a metal layer (for example gold or silver) on both sides, each of which acts as an electrode. It has been known for some time that these composites can act as electrochemical transducers that behave like artificial muscles for biomechanical and

biomimetic applications. Their operation requires humidification because the ion conduction in the polymer typically requires swelling using a solvent, which is usually water. One of the obstacles to the widespread use of such transducers is thus their dependence on the degree of hydration, as evaporation becomes a potential issue. The narrow electrochemical stability window of water also limits the extent of the applied voltage and, thus, the level of strain that the actuators can generate.

There is considerably more scope for production of high levels of strain in the intrinsically conducting polymer variant of these devices. Electrode materials, such as polypyrrole or polyaniline, are used that show a change in volume as they are cycled between their oxidized and reduced states. Aqueous or aprotic solvent electrolytes have conventionally been used, but these limit the performance and lifetime of the devices for the reasons discussed above. The issue may be solved by turning to ionic liquids. The use of ionic liquids in electrochemical actuators was pioneered by Lu and co-workers^{55,56}, who showed that ionic liquids could indeed provide a huge increase in the performance of these devices. Owing to the use of ionic liquids, an operational life in excess of 10^6 cycles is now an easily achievable target.

Since the beginning of these studies, it has been clear that the origin of the improvement was associated with the high electrochemical stability of the ionic liquids, discussed above, which allows the conducting polymer to undergo its redox cycle with no interference from side reactions involving the electrolyte solvent. This prompted intensive further work in this field over the past five years. For example, Zhou *et al.*⁵⁷ demonstrated the fabrication of a completely solid-state device. This was achieved by gelling the electrolyte by inclusion of a cross-linked polymer component, which was polymerized *in situ*. An additional benefit of the use of hydrophobic ionic liquids is that they have less tendency to absorb or mix with water in an aqueous environment, thus enabling potential *in vivo* applications. Xi *et al.*⁵⁸ have shown that the use of an ionic liquid electrolyte allowed a poly(3-methylthiophene)-based actuator to achieve increased isotonic actuation with increased stress. Hence, the work output per cycle increases with increasing stress, a highly welcome feature for a practical device. Lu *et al.*⁵⁹ have described axial-geometry actuators based on polyaniline yarn and hollow fibres; these actuators were able to generate stresses of up to 0.85 MPa, exceeding that of skeletal muscle. The 'yarn-in-fibre' approach has the potential to allow the creation of a multi-fibre device in which the required stress capability can be achieved by simple combination of a sufficient number of fibres.

The other feature of ionic liquids as electrolytes for use in combination with conducting polymers is that the anions typically involved, such as PF_6^- , BF_4^- , CF_3SO_3^- or $[(\text{CF}_3\text{SO}_2)_2\text{N}]^-$, are often the same as those used as counter ions in the conducting polymers. The reason for this is related to the similar set of properties that are optimum in both cases: diffuse or shielded charge creating a weakly interacting ion of low basicity⁶⁰. Thus, the active ion is present in the ionic-liquid electrolyte at high concentration, often around 5 mol l^{-1} , and the redox cycling process of such polymer-electrolyte combinations takes place without any diffusion limit in the electrolyte. This feature also allows the polymerization of the conducting polymer to be carried out in the ionic liquid⁶¹. Carbon-nanotube-based actuators with ionic liquids as electrolytes have recently been described^{62,63}.

In most actuator applications, it is important that the ionic liquid is present in a gelled or otherwise semi-solid form, while still maintaining its ion conduction properties. A number of approaches have been investigated to achieve this, with some degree of success. Vidal and co-workers have developed ionic-liquid-based interpenetrating network electrolyte systems for use in actuators and shown that these can produce very long lifetimes^{64–66}. An alternative approach has involved the use of nitrile rubber impregnated with

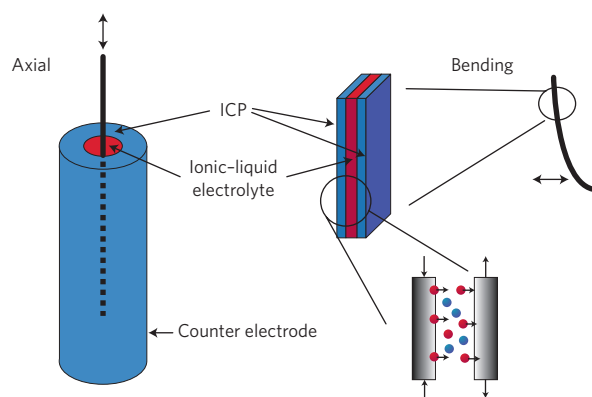


Figure 8 | An electromechanical actuator produces a mechanical bending or axial motion in response to an electrical stimulus. The voltage applied to the two intrinsically conducting polymer (ICP) electrodes causes a small oxidation or reduction reaction to take place, changing the volume of the electrode material. This produces a stress in the material at usefully high strain. Gelled ionic-liquid-based electrolytes provide the ideal combination of high conductivity coupled with high thermal and electrochemical stability for this technology.

an ionic liquid^{67,68}. In this case, the actuator displacement was found to vary with anion size, as expected if the change in volume of the actuator is principally the result of incorporation of the anion. A further approach that is gaining some attention is the possible use of organic ionic plastic crystal solid electrolytes^{69,70}. Devices utilizing membranes into which ionic liquids have been impregnated have also been reported^{71,72}; these show considerably improved stability over aqueous electrolyte analogues.

Overall, the prospects appear to be excellent for practical biomimetic actuator devices having the required strain, lifetime and rate capabilities. For reasons associated with both properties and chemical structure, ionic liquids have become the electrolyte materials of choice and are contributing to significant progress in this field.

Concluding remarks

Lewis-neutral ionic liquids date from just 15 years ago, and have been the basis of a quiet revolution in materials science; their use is increasingly being implemented in devices. In 2007, a conference on the subject drew the attendance of several hundred scientists. We have focused here on the applications of ionic liquids in the field of energy, which is today's greatest challenge because of its implications in climate change, war, peace and humanitarian progress. It is clear that the field of batteries is very close to maturity for transportation purposes, and, as stated, safety is the main hurdle that ionic liquids can help to overcome. Conventional fuel cells have received enormous funding and publicity but have returned disappointing results. Here, a change of direction is needed. This may occur perhaps through the use of a Grotthuss conduction mechanism for the proton but not one based on water. Ultimately, especially if the 'hydrogen economy' proves too difficult to implement, biofuel cells in which enzymes can break down more complex molecules like ethanol will be required. The ability of the ionic liquid to stabilize the complex protein folding that allows the activity of an enzyme is, in itself, a revolution.

In the past two or three years, the tremendous potential offered by ionic liquids as solvents and electrolytes has been realized in a broad swath of materials preparation and device applications⁷³. Ionic liquids are thus no longer laboratory curiosities; they have become an important component in the materials science field.

Nonetheless, there remain many aspects of the behaviour of ionic liquids that need urgent investigation if the applications discussed here are to be better understood and developed. Examples

are fundamental electrochemical issues such as the structure of the double layer at an electrode in an ionic liquid; the speciation of solute ions such as metal ions; transference numbers and how these are influenced by speciation; dielectric properties and how these merge with conduction properties as a function of frequency; surface chemistry and wetting phenomena of ionic liquids in porous electrodes; thermodynamic quantities such as chemical potentials of solutes in ionic liquids, and how these influence redox potentials; ion association and its effect on thermodynamic and transport properties; and interactions of ionic liquids with solutes and interfaces. The simple dissolution of a lithium salt always results in electrolytes in which only a fraction of the current is actually carried by Li^+ ions. It would be interesting to see whether the incorporation of poly(Li salts), that is, lithium salts with a polyanionic chain, through specific interaction with the fixed negative charge of polarizing Li^+ ions, could change the flux balance.

All of these fundamental aspects of the physical chemistry and electrochemistry of ionic liquids remain to be thoroughly explained, and they promise to further improve the potential of their various electrochemical applications.

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